

CURRICULUM VITAE



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Research Interests:

- Neo-virus lifestyle of fungal viruses.
- Cross-kingdom virus transmissions between plants and fungi.
- Symbiotic relationships between fungi and viruses.
- Development of CRISPR-based point-of-care (POC) diagnostics for viruses.

Education:

- Ph.D. (2009-2013) in **Molecular Biology**, Lincoln University, New Zealand.
- M.Sc. (2003-2005) in **Genetics**, University of Calcutta, India.
- B.Sc. (2000-2003) in **Microbiology** (honors), University of Calcutta, India.

Work Experience:

- **Senior Scientist**, Virology and Phytoplasmology team, Plant Health & Environment Laboratory, Ministry for Primary Industries, **New Zealand** (Apr 2023 to Present).
- **Postdoctoral Research Associate**, Animal Disease Research & Diagnostic Laboratory, Veterinary & Biomedical Sciences, South Dakota State University, **United States of America** (Nov 2021 to March 2023).
- **Assistant Professor** (specially appointed), Institute of plant science and resources (IPSR), Okayama University, **Japan** (May 2020 to Mar 2021).
- **JSPS Postdoctoral Fellow**, Institute of plant science and resources (IPSR), Okayama University, **Japan** (May 2018 to May 2020).
- **SERB (DST) Young Scientist**, Division of plant biology, Bose Institute, **India** (Apr 2015 to Mar 2018).
- **Honorary Guest Worker**, Division of plant biology, Bose Institute, **India** (Nov 2014 to Mar 2015).
- **Researcher**, Plant virus laboratory and biotechnology unit, Central Research Institute for Jute and Allied Fibres (CRIJAF), **India** (Jul 2008 to Jan 2009).
- **Senior Research Fellow**, Plant virus laboratory and biotechnology unit, Central Research Institute for Jute and Allied Fibres (CRIJAF), **India** (Feb 2007 to Jul 2008).
- **Junior Research Fellow**, Manovikas biomedical research and diagnostic centre, **India** (Dec 2005 to Jan 2007).

Honors & Awards:

- **María Zambrano Postdoctoral fellowship** for attracting international talent (at Universidad Politécnica De Madrid), Spain, 2022 (Respectfully declined).
- **Excellence Prize** for an outstanding academic oral presentation at the “Virology Meeting for Early Career Researchers (Virus wakate 2020)”, Japan, 2020.
- **Foreigner Young Scientist Grant**, Ministry of Science and Technology of the People’s Republic of China, China, 2020 (Respectfully declined).
- **Japan Society for the Promotion of Science (JSPS) Postdoctoral Fellowship** (Standard), Japan, 2018.
- **SERB Start-up Research Grant under Young Scientist scheme**, Science and Engineering Research Board (SERB), India, 2014.
- **IVS-NIV travel bursary award**, Indian Virological Society (IVS), India, 2013.
- **Bio-Protection Research Writing Scholarship**, Lincoln University, New Zealand, 2013.
- **PhD fellowship** at Lincoln University, New Zealand, 2009.
- **Senior Research Fellowship (SRF)** at Central Research Institute for Jute and Allied Fibres (CRIJAF), India, 2007.
- **Junior Research Fellowship (JRF)** at Manovikas Biomedical Research and Diagnostic Centre, India, 2005.

Memberships:

Full member of the American Society for Virology (ASV) since January 2024.

Research projects involved in:

Ongoing project in New Zealand (as a senior scientist):

“PLANT-dx 2.0 – Rapid, Simple and Scalable Plant Pathogen Detection”. *Host Institution in New Zealand:* Plant Health and Environment Laboratory (PHEL), Ministry for Primary Industries (MPI), Auckland. *Funded by:* Bill & Melinda Gates foundation (Investment ID: INV-038694). *Funding amount:* **USD 1 million** for two years (2023-2025). **Prof. Julius B. Lucks** (Northwestern University, USA) is the PI of this collaborative project and Dr. Jeremy Thompson is Subha Das’s manager at MPI.

Completed in India (as a Principal Investigator):

“Molecular characterization of double-stranded RNA mycoviruses potentially affecting virulence of *Rhizoctonia solani* (AG-1-IA) causing sheath blight disease of rice in India”. *Host Institution:* Bose Institute (India), *Funded by:* Science and Engineering Research Board (SERB, sanction Order no.: SB/YS/LS-68/2014, Department of Science & Technology, Government of India) under the start-up research grant under “**Young Scientist**” scheme. *Funding amount:* **INR 3.25 million (~USD 39146)** for three years (2015-2018).

Completed in Japan (as a JSPS Postdoctoral Fellow):

“Proof of concept in the new lifestyle (yado-kari nature) of yado-kari virus”. *Host institution:* The Institute of Plant Science and Resources (IPSR), Okayama University (Japan), *Funded by:* Japan Society for the Promotion of Science (JSPS) for two years (2018-2020) under the “**JSPS Postdoctoral Fellowship**” (standard) for Research in Japan (Fellowship ID: P18086), *Host researcher:* Prof. Nobuhiro Suzuki. *Grant-in-aid:* **JPY 2.4 million (~USD 16868)** for two years.

Approved in China (as a Foreigner Young Scientist):

“The mechanism of circular non-coding small RNA-attenuated pathogenicity of phytopathogenic fungi”. *Host institution*: The College of Plant Protection, Northwest A&F University (China), *Funded by* the Ministry of Science and Technology of the People’s Republic of China under the “**Foreigner Young Scientist Grant**” scheme (Grant number: QN20200027004, duration Jan 1st, 2021- Dec 31st, 2022), *Host researcher*: Prof. Liying Sun. *Grant amount*: **RMB 0.22 million (~USD 30152)**. *Did not accept the offer due to Covid-19 pandemic*.

Scientific publications:

1. Sato Y, **Das S**, Velasco L, Turina Massimo, Osaki H, Kotta-Loizou I, Coutts RHA, Kondo H, Sabanadzovic S, Suzuki N (2023), ICTV Virus Taxonomy Profile: Yadokariviridae 2023. **Journal of General Virology** 104:001826; DOI: 10.1099/jgv.0.001826.
2. Workman AM, McDanel TG, Harhay G, **Das S**, Loy JD, Hause BM (2022). Recent emergence of bovine coronavirus variants with mutations in the hemagglutinin-esterase receptor binding domain in U.S. cattle. **Viruses** 14(10), 2125; <https://doi.org/10.3390/v14102125>
3. **Das S**, Hisano S, Eusebio-Cope A, Kondo H, Suzuki N (2022). A transfectable fusagravivirus from a Japanese strain of *Cryphonectria carpnicola* with spherical particles. **Viruses** 14(8), 1722; <https://doi.org/10.3390/v14081722>.
4. **Das S**, Alam MM, Zhang R, Hisano S, Suzuki N (2021). Proof of concept for the yadokari nature: a capsidless replicase-encoding but replication-dependent (+)ssRNA virus hosted by an unrelated dsRNA virus. **Journal of Virology** 95(17), e00467-21. DOI: <https://doi.org/10.1128/JVI.00467-21>
5. Neang S, Bincader S, Rangsuwan S, Keawmanee P, Rin S, Salaipeth L, **Das S**, Kondo H, Suzuki N, Sato I, Takemoto D, Rattanakreetakul C, Pongpisutta R, Arakawa M, Chiba S (2021). Omnipresence of partitiviruses in rice aggregate sheath spot symptom-associated fungal isolates from paddies in Thailand. **Viruses** 13(11), 2269; <https://doi.org/10.3390/v13112269>.
6. Lian Z, **Das S**, Luo J, Andika IB, Sun L (2021). Complete genome sequence of a novel ourmia-like mycovirus infecting the phytopathogenic fungus *Botryosphaeria dothidea*. **Archives of Virology** 166, 3461-3465 (**equal contribution**).
7. **Das, S** and Suzuki, N. Yado-kari virus 1 and yado-nushi virus 1 (Unassigned). Editor(s): Dennis H. Bamford, Mark Zuckerman, **Encyclopedia of Virology** (Fourth Edition), Academic Press, 2021, Pages 658-663, ISBN 9780128145166. DOI: [10.1016/B978-0-12-809633-8.20949-7](https://doi.org/10.1016/B978-0-12-809633-8.20949-7)
8. **Das S***, Das S (2019). First report of a novel alphapartitivirus in the basidiomycete *Rhizoctonia oryzae-sativae*. **Archives of Virology** 164, 889-892.
9. **Das S***, Falloon RE, Stewart A, Pitman AR (2016). Novel mitoviruses in *Rhizoctonia solani* AG-3PT infecting potato. **Fungal Biology** 120, 338-350.
10. **Das S***, Falloon RE, Stewart A, Pitman AR (2014). Molecular characterisation of an endornavirus from *Rhizoctonia solani* AG-3PT infecting potato. **Fungal Biology** 118, 924-934.
11. **Das S**, Shah FA, Butler RC, Falloon RE, Stewart A, Raikar S, Pitman AR (2014). Genetic variability and pathogenicity of *Rhizoctonia solani* associated with black scurf of potato in New Zealand. **Plant Pathology** 63, 651-666.
12. Roy A, Acharyya S, **Das S**, Ghosh R, Paul S, Srivastava RK, Ghosh SK (2009). Distribution, epidemiology and molecular variability of the begomovirus complexes associated with yellow vein mosaic disease of mesta in India. **Virus Research** 141, 237-246.

13. Paul S, Ghosh R, **Das S**, Palit P, Acharyya S, Das A, Mir JI, Chaudhuri S, Ghosh SK, Roy A (2009). First report of *Tomato leaf curl Joydebpur virus* and associated betasatellite in kenaf (*Hibiscus cannabinus*) plants showing leaf curl symptoms from southern India. **Plant Pathology** 58, 403.
14. **Das S**, Roy A, Ghosh R, Paul S, Acharyya S, Ghosh SK (2008). Sequence variability and phylogenetic relationship of betasatellite isolates associated with yellow vein mosaic disease of mesta in India. **Virus Genes** 37, 414-424.
15. **Das S**, Ghosh R, Paul S, Roy A, Ghosh SK (2008). Complete nucleotide sequence of a monopartite begomovirus associated with yellow vein mosaic disease of mesta from north India. **Archives of Virology** 153, 1791-1796.
16. Paul S, Roy A, Ghosh R, **Das S**, Chaudhuri S, Ghosh SK (2008). Molecular characterisation and sequence variability of betasatellites associated with leaf curl disease of kenaf (*Hibiscus cannabinus* L.) from different geographical locations of India. **Acta Virologica** 52, 251-256.
17. Ghosh R, Paul S, **Das S**, Palit P, Acharyya S, Das A, Mir JI, Ghosh SK, Roy A (2008). Molecular evidence for existence of a New World begomovirus associated with yellow mosaic disease of *Corchorus capsularis* in India. **Australasian Plant Disease Notes** 3, 59-62.
18. Ghosh R, Paul S, Palit P, **Das S**, Das A, Acharyya S, Mir JI, Ghosh SK, Roy A (2008). Molecular characterisation and computational analysis of coat protein gene of an East Indian isolate of *Bhendi yellow vein mosaic virus* and associated DNA-beta satellite. **Indian Journal of Virology** (now **VirusDisease**) 19, 173-182.
19. Dutta S, **Das S**, Guhathakurta S, Sen B, Sinha S, Chatterjee A, Ghosh S, Ahmed S, Ghosh S, Usha R. (2007). Glutamate receptor 6 Gene (Glur6 or GRIK2) polymorphisms in the Indian population: a genetic association study on autism spectrum disorder. **Cellular and Molecular Neurobiology** 27, 1035-1047.

Research Presentations:

1. The 43rd Annual meeting of the American Society for Virology, ASV 2024 (June 2024), Columbus, Ohio, USA (**Poster presentation**).
2. Virology meeting for early career researchers 2020 (Dec 2020), Japan (**Oral presentation** at online Zoom meeting).
3. The 39th Annual meeting of the American Society for Virology, ASV 2020 (Jun 2020), Colorado, USA (Selected for **oral presentation**, meeting cancelled).
4. Asian Mycological Congress 2019 (Oct 2019), Mie, Japan (**Oral presentation**).
5. International neovirology mini-symposium series IV (Feb 2019), Kurashiki, Japan (**Oral presentation**).
6. 26th National conference of Indian Virological Society, VIROCON 2017 (Dec 2017), Mangaluru, India (**Oral presentation**).
7. International conference on “Global Perspectives in Virus Disease Management”, VIROCON 2016 (Dec 2016), Bengaluru, India (**Poster presentation**).
8. National symposium on “Evolving plant biology: from chromosomes to genomics” (Nov 2014), Kolkata, India (**Poster presentation**).
9. Asia-Pacific Congress of Virology, VIROCON-2013 (Dec 2013), Noida, India (**Oral presentation**).
10. Potatoes New Zealand R&D investment meeting (Oct 2013), Wellington, New Zealand (**Oral presentation**).
11. Bio-Protection symposium on “Managing pests: the future of biocontrol” (Aug 2011), Christchurch, New Zealand (**Poster presentation**).
12. 6th Australasian soilborne diseases symposium (Aug 2010), Queensland, Australia (**Oral presentation**).

13. International conference on “Emerging and re-emerging viral diseases of the tropics and subtropics” (Dec 2007), New Delhi, India (**Poster presentation**).
14. National symposium on “Microbial diversity and plant health” (Nov 2007), Kalyani, India (**Poster presentation**).

Invited talks:

1. “Microbes against microbes: an eco-friendly way of managing fungal plant pathogens”, Tokushima Prefectural Tomioka Higashi High School, Tokushima, Japan (Dec 14, 2018). Invited by the **Japan Society for the Promotion of Science (JSPS)** under “**JSPS Science Dialogue Program**”.

Invited attendee in an international symposium:

1. Invited to act as a moderator to host an open discussion on “How to enjoy academic life for early career researchers” at the “**International Plant Web Forum 2021**” organized by the Institute of Plant Science and Resources, Okayama University, Japan (Sep 6-7, 2021).

Conference abstracts:

1. **Das S**, Demissie H, Dreyer K, Leonard J, Jewett M, Mangan N, Dida M, Nkalubo ST, Lucks J, Thompson JR (2024). An inexpensive robust point-of-use virus diagnostic platform based on NASBA-Cas13a. The official abstracts of the IUMS 2024 will be published by Frontiers supported by the Frontiers portfolio of Microbiology journals, abstract no. 810.
2. **Das S**, Demissie H, Dreyer K, Leonard J, Jewett M, Mangan N, Dida M, Lucks J, Thompson JR (2024). A robust and sensitive plant virus detection platform based on NASBA-Cas13a. Scientific program and abstracts of 43rd annual meeting of American Society for Virology (ASV 2024), p 555.
3. **Das S** (2020), Proof of concept for the neo-viral lifestyle of yado-kari virus 1 being hosted by yado-nushi virus 1. Virology Meeting for Early Career Researchers 2020 (Japan) meeting abstract book, p 13.
4. **Das S**, Alam M, Zhang R, Hisano S, Sato Y, Kondo H, Suzuki N (2020). 2A-mediated co-translational cleavage is prerequisite for the viability of yado-kari virus 1 hosted by yado-nushi virus 1. Scientific program and abstracts of 39th annual meeting of American Society for Virology (ASV 2020), p 378-379.
5. **Das S**, Alam M, Zhang R, Hisano S, Sato Y, Kondo H, Suzuki N (2019). 2A-like protease activity is essential for replication and viability of yado-kari virus 1 hosted by yado-nushi virus 1. Asian mycological congress 2019. Identification number: 3-O20-2.
6. **Das S**, Das S (2017). Mycoviral diversity in *Rhizoctonia solani* AG-1-IA infecting rice in India. 26th National conference of Indian virological society, VIROCON 2017, p 103.
7. **Das S**, Das S (2016). Identification and characterisation of mycoviruses in *Rhizoctonia solani* AG-1-IA associated with sheath blight diseases of rice in India. International conference on global perspectives in virus disease management, VIROCON 2016, p 157.
8. **Das S**, Falloon RE, Pitman AR (2014). Endornaviruses are ubiquitous in *Rhizoctonia solani* infecting potato. National symposium on evolving plant biology: from chromosomes to genomics, p 52.
9. **Das S**, Falloon RE, Pitman AR (2013). Characterisation of mycoviruses in *Rhizoctonia solani* infecting potato. Asia-Pacific congress of virology, VIROCON 2013, p 116.
10. **Das S**, Shah FA, Falloon RE, Butler RC, Pitman AR (2010). Characterisation of *Rhizoctonia solani* anastomosis group 2-1 from potato tubers in New Zealand. 6th Australasian soilborne diseases symposium, p 41.

11. Roy A, Ghosh R, Paul S, **Das S**, Palit P, Acharyya S, Das A, Mir JI, Ghosh SK (2008). Molecular evidence for existence of new emerging begomoviruses from bast fibre crops in India. *Indian Journal of Virology* 19, 76-77.
12. Das A, Paul S, Ghosh R, **Das S**, Palit P, Acharyya S, Mir JI, Ghosh SK, Roy A (2008). Association of a *Begomovirus* with yellow mosaic disease of *Clerodendron* sp. occurring in Eastern India as revealed by *In-silico* analysis. *Indian Journal of Virology* 19, 118.
13. Ghosh R, Paul S, **Das S**, Acharyya S, Das A, Palit P, Mir JI, Ghosh SK, Roy A (2008). First report of association of a satellite DNA from croton plant showing yellow vein symptoms in Eastern India. *Indian Journal of Virology* 19, 118.
14. **Das S**, Paul S, Ghosh R, Palit P, Acharyya S, Das A, Mir JI, Ghosh SK, Roy A (2008). Occurrence of a *Begomovirus* and satellite DNA associated with yellow vein disease of *Sida* sp. from eastern India and its computational analyses. *Indian Journal of Virology* 19, 119.
15. Acharyya S, Paul S, Ghosh R, **Das S**, Palit P, Das A, Mir JI, Ghosh SK, Roy A (2008). Molecular evidence for natural occurrence of *Bhendi yellow vein mosaic virus* in *Urena lobata* from Eastern India. *Indian Journal of Virology* 19, 119.
16. Palit P, Ghosh R, Paul S, **Das S**, Das A, Acharyya S, Mir JI, Ghosh SK, Roy A (2008). Characterisation of coat protein gene of *Bhendi yellow vein mosaic virus* and associated satellite DNA from Eastern India. *Indian Journal of Virology* 19, 119-120.
17. Paul S, Ghosh R, **Das S**, Acharyya S, Palit P, Das A, Mir JI, Ghosh SK, Roy A (2008). Detection of a *Begomovirus* associated with yellow vein disease of *Eclipta prostrata* in eastern India. *Indian Journal of Virology* 19, 120.
18. Ghosh R, Paul S, **Das S**, Das A, Acharyya S, Ghosh SK, Roy A (2008). Molecular detection of a new world bipartite begomovirus in jute crop. XVIII national conference of Indian virological society, p 44.
19. Das A, Ghosh R, Paul S, **Das S**, Acharyya S, Ghosh SK, Roy A (2008). Sequenced based evidence for occurrence of *Begomovirus* and satellite DNA- β associated with leaf deformity syndrome of sunn hemp in East India. A journey from plant physiology to plant biology. An international symposium to commemorate the 150th birth anniversary of Sir J. C. Bose & birth centenary of Prof. S.M. Sirkar, p 70.
20. Ghosh R, Paul S, **Das S**, Das A, Acharyya S, Ghosh SK, Roy A (2008). Detection and characterisation of a new world begomovirus in jute (*Corchorus capsularis*) from India. Abstract book of "A journey from plant physiology to plant biology". An international symposium to commemorate the 150th birth anniversary of Sir J. C. Bose & birth centenary of Prof. S.M. Sirkar, p 82.
21. Roy A, Ghosh R, Paul S, **Das S**, Palit P, Acharyya S, Das A, Mir JI, Ghosh SK (2008). Newly emerging begomoviruses on bast fibre crops in India. International symposium on jute and allied fibres production, utilisation and marketing, p 83.
22. **Das S**, Ghosh R, Paul S, Palit P, Acharyya S, Das A, Mir JI, Ghosh SK, Roy A (2007). Molecular variability of *Begomovirus* complex associated with yellow vein mosaic disease of mesta in Northern and Eastern India. National symposium on microbial diversity and plant health, P 11.
23. Roy A, Ghosh R, Paul S, **Das S**, Palit P, Acharyya S, Mir JI, Ghosh SK (2007). Characterisation of a monopartite recombinant begomovirus and satellite DNA β associated with yellow vein mosaic disease of mesta crop in India. 10th plant virus epidemiology symposium, p 59.
24. Ghosh R, Paul S, **Das S**, Palit P, Acharyya S, Mir JI, Roy A, Ghosh SK (2007). Association of *Begomovirus* complex with yellow mosaic disease of jute in India. 10th plant virus epidemiology symposium, p 87.
25. Paul S, Ghosh R, **Das S**, Palit P, Acharyya S, Mir JI, Chaudhary S, Roy A, Ghosh SK (2007). First sequence based evidence for occurrence of recombinant *Begomovirus* and a satellite β -DNA associated with leaf curl disease of kenaf (*Hibiscus cannabinus*) in Northern India. 10th plant virus epidemiology symposium, p 118.

Peer Review Activities:

Reviewer for scientific journals:

1. *Archives of Virology*, ISSN: 1432-8798.
2. *Archives of Microbiology*, ISSN: 0302-8933.
3. *Viruses*, ISSN: 1999-4915
4. *Virus Research*, ISSN: 0168-1702
5. *International Journal of Molecular Sciences*, ISSN: 1422-0067.

List of referees:

- **Dr Jeremy Thompson**
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- **Prof. Nobuhiro Suzuki**
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The information provided above is correct to the best of my knowledge.

Place: Auckland, New Zealand

Date: 08/02/2025



(Subha Das)